

Capstone Final Report

Natural Scene Classification from Photographs

1. Problem Statement

Can a model automatically classify a photograph into one of six natural scene categories — buildings, forest, glacier, mountain, sea, or street — based on its visual content alone? This project builds and compares three classification approaches on the Intel Image Classification dataset (originally created for an Intel/Analytics Vidhya competition), a benchmark of roughly 17,000 labeled 150x150 photographs.

Automated scene recognition has direct commercial relevance: photo management apps use it to auto-organize libraries, mapping and navigation systems use it to interpret street-level or satellite imagery, and content platforms use it to tag or moderate uploaded images at scale. The natural client framing for this project is a product team at a photo-management or travel platform (e.g. a Google Photos-style auto-tagging feature team) looking to automatically categorize a user's photo library by scene type.

2. Data and Approach

The dataset was split into 11,928 training images, 2,106 validation images, and 3,000 held-out test images, with a roughly even distribution across the six classes (see Table 1 in the accompanying Model Metrics file for exact per-class training counts).

Rather than jumping straight to the most powerful available technique, three models of increasing complexity were built and compared, so the value each added could be measured directly:

- Random Forest on 7 hand-engineered image features (mean RGB, brightness, contrast, edge density, and a blue-hue fraction extracted via HSV thresholding) — a fast, interpretable classical ML baseline.
- A convolutional neural network (CNN) trained from scratch on raw, normalized pixels — letting the model learn its own visual features rather than relying on the 7 engineered ones.
- Transfer learning using a MobileNetV2 backbone pretrained on ImageNet, with a new classification head trained on top of the frozen, pretrained features.

All three models were trained on the same training split and compared on the same validation split. The model with the best validation performance was then evaluated exactly once on the untouched test set, to produce an honest final performance estimate.

3. Findings

3.1 Model Comparison

The three models showed a clear, large performance gap, summarized below (full hyperparameters and per-class metrics are documented separately in the Model Metrics file):

Model	Val. Accuracy	Val. Macro F1
Transfer Learning (MobileNetV2)	90.1%	0.903
CNN (from scratch)	81.3%	0.814
Random Forest (engineered features)	53.5%	0.524

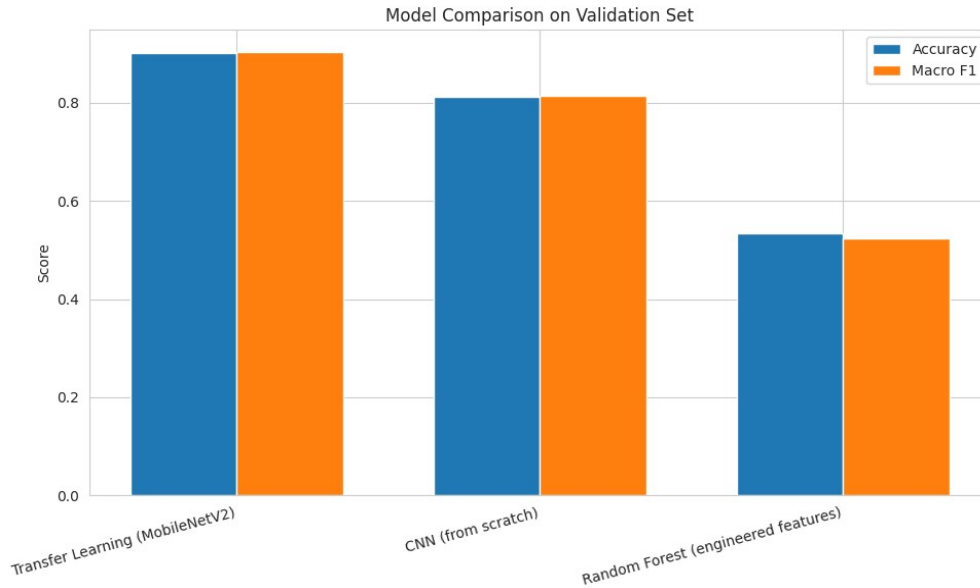


Figure 1. Validation accuracy and macro F1 across all three models.

The Random Forest baseline, built on just 7 simple numeric features, was clearly outmatched — confirming that a handful of aggregate color/edge statistics cannot capture the spatial patterns that actually distinguish these scenes. The from-scratch CNN improved substantially (81.3% accuracy) by learning its own visual features directly from pixels, but was still clearly outperformed by transfer learning.

Transfer learning with MobileNetV2 was the strongest model by a wide margin, reaching 90.1% validation accuracy. This is the expected pattern when a dataset (here, ~12,000 training images) is small relative to what a CNN needs to learn strong visual features entirely from scratch: reusing features already learned from ImageNet's 1.4 million images gave the model a substantial head start.

3.2 Final Model Performance on the Test Set

The selected final model (MobileNetV2 transfer learning) was evaluated once on the 3,000-image held-out test set, confirming the validation-set result was not overly optimistic:

- Test accuracy: 90.6%
- Test macro F1: 0.909

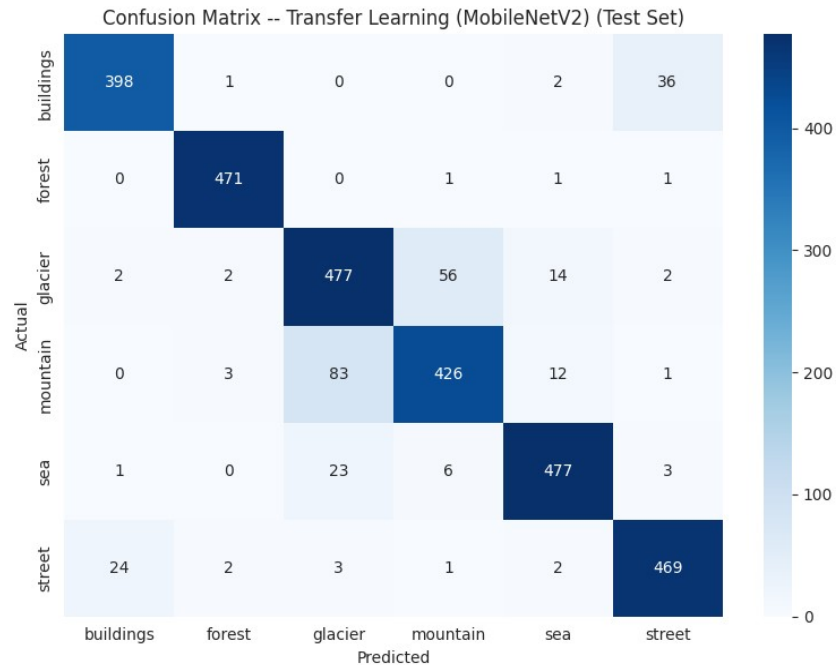


Figure 2. Confusion matrix for the final model on the test set.

Predictions were highly accurate for forest (99% F1) and buildings/sea/street (92–94% F1). The model's errors were concentrated almost entirely in one place: confusing glacier and mountain images with each other (56 glacier images predicted as mountain; 83 mountain images predicted as glacier). This is a sensible, explainable failure mode — glacier and mountain photographs frequently share the same rocky, snow-covered visual palette and composition, and even a human glancing quickly at a thumbnail could plausibly make the same mistake. A smaller, secondary confusion appeared between buildings and street (36 and 24 images respectively), which similarly share overlapping urban color palettes and structural patterns.

3.3 What the Random Forest Baseline Reveals

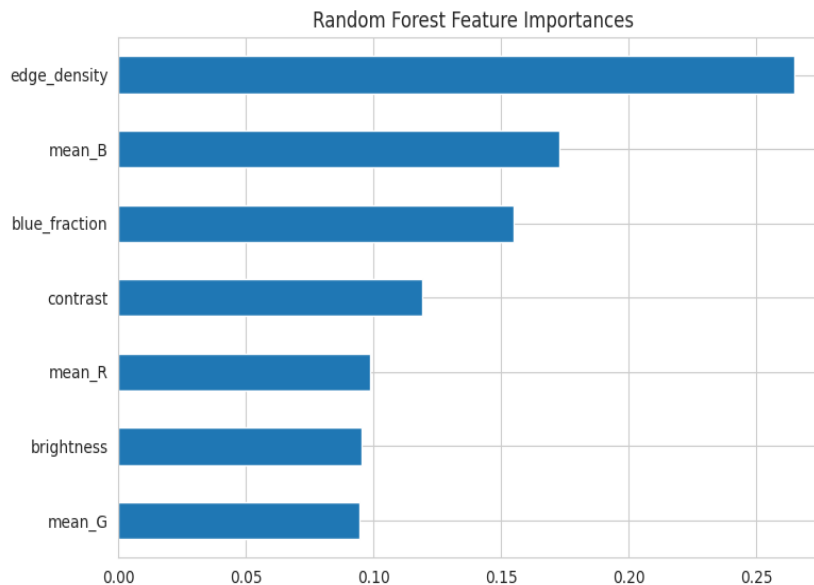


Figure 3. Random Forest feature importances, engineered-feature baseline.

Even though the Random Forest baseline was the weakest model overall, its feature importances are informative on their own: they show which simple visual cues (e.g. color balance, edge density) carry the most signal about scene type, providing an interpretable sanity check on what the more complex models should plausibly be picking up on as well.

4. Recommendations

- 1. Deploy the MobileNetV2 transfer-learning model for production auto-tagging, not the from-scratch CNN or the engineered-feature baseline. At 90.6% test accuracy, it is accurate enough to meaningfully reduce manual tagging effort, while remaining lightweight enough (a frozen, mobile-optimized backbone) to run efficiently even on-device or at scale.
- 2. Prioritize additional labeled data or targeted data augmentation specifically for glacier and mountain images, since that pairing accounts for the large majority of the model's remaining errors. A relatively small, targeted data collection effort focused on this one confusion could meaningfully raise overall accuracy, more efficiently than collecting more data uniformly across all six classes.
- 3. If the deployed system needs to flag uncertain predictions for human review (e.g. in a semi-automated tagging pipeline), use the model's per-class prediction confidence as a review trigger specifically for the glacier/mountain boundary — since that is precisely where the model's confidence is least reliable, a targeted review step there will catch the bulk of likely errors without requiring review of every prediction.

5. Ideas for Further Research

- Fine-tuning: unfreezing the top layers of the MobileNetV2 backbone and fine-tuning them on this dataset (rather than only training a new classification head) would likely improve accuracy further, particularly on the glacier/mountain confusion, at the cost of longer training time.
- Data augmentation: systematic augmentation (random crops, rotations, brightness/contrast jitter) during CNN and transfer-learning training was not explored in this iteration, and could improve generalization, especially for the from-scratch CNN.
- Alternative pretrained backbones: comparing MobileNetV2 against other architectures (e.g. EfficientNet, ResNet) pretrained on ImageNet could reveal whether a larger backbone trades off accuracy against the inference-speed benefits MobileNetV2 is specifically designed for — relevant if the deployment target is mobile or edge devices.
- Error analysis at the image level: manually reviewing a sample of the misclassified glacier/mountain images could reveal whether errors cluster around a specific sub-type (e.g. distant mountain peaks with snow, which visually resemble glaciers) that could be addressed with a more targeted modeling or data strategy.